Robot Design Speech

Write it in your own words. Facts, numbers and reasons.

Updated 2026-11-12. The blue handwriting-style text is what the team wrote on 2026-11-05. Each section has a review box that says whether it is at Developing, Accomplished or Exceeds so far. **16 days to the tournament.**

Name: _____ **Date:** _____

At judging we give a **5-minute robot design explanation** with no game field, then judges ask questions for 5 more minutes. These pages guide you through each part of the speech. They give you sentence starters, not the answers. **You** write what we really did, so it sounds like us.

What to do

1. **Read** the gold box on each page. It tells you what judges listen for.
2. **Write** an answer under each ✎ starter. Use the reference pages (meeting notes, design log, test results) to find facts.
3. **Add numbers.** How many runs? How far off? How many out of 10?
4. **Don't know?** Write **[ASK: name]** and we will track it down.
5. **Practice out loud.** About 600 to 700 words is 5 minutes. Speak slowly.

What makes a speech score high?

- **Say why.** "We changed it because ..." beats "We changed it."
- **Say what is new.** Tell the judges what we changed from the original build.
- **Tell the test story:** we tested, found a problem, changed it, tested again, it got better.
- **Name everyone.** All nine of us speak and all nine of us did real work.
- **Be proud.** Smile. Celebrate what we did.

The robot design process we follow

Identify → Design → Create → Iterate (test and improve) → Communicate what we learned. The seven speech sections follow this order.

What judges listen for

Judges tick one box per row. Clear evidence is **Accomplished (3)**. To score **Exceeds (4)** the judges must write down how we exceeded, so give them something to write.

Area	Accomplished: clear evidence of ...	Exceeds: give them more
IDENTIFY	Mission strategy. Use of building and coding resources.	Why we chose missions in that order. Which resources, and what we took from each. Who we asked for help.
DESIGN	All members contributed ideas. Building and coding skills in all members.	Name **every** teammate and one thing each did. Who learned a new skill?
CREATE	Clear explanation of attachments and their purpose. Clear explanation of code and sensors.	What we **changed from the original** and why. Code explained **step by step**. What each sensor does for us.
ITERATE	Repeated testing. Improvements based on testing.	**Numbers** (how many runs, how far off). A story: test, problem, change, test again.
COMMUNICATE	Detailed process and lessons learned. Pride and enthusiasm.	Advice for another team. A mistake we learned from. Smiles and teamwork.

Who says what

Everyone speaks: Sylvie, Brynn, Jack, Fedor, Ira, Jordan, Ella, Alexis, Amelia. Times add up to about 4 minutes 40 seconds. You can split a section between two people.

Section	Time	Speaker(s)
1. Opening and mission strategy	0:30	
2. Choosing and building our robot	0:40	
3. Our attachments	1:00	
4. Our code and sensors	0:50	
5. Testing and improving	0:50	
6. What we learned	0:25	
7. Teamwork and closing	0:25	

Where our speech is now

Levels follow the rubric: Beginning (1), Developing (2), **Accomplished (3)**, Exceeds (4). This is a review of what is written so far, not a score from judges.

Area	Level now	What we have	To go up a level
IDENTIFY	DEVELOPING	Reasons for the missions and their order. Resources named.	Name the missions. Say what we took from each resource and who helped.
DESIGN	DEVELOPING	Four teammates named in the forklift story. Squaring shows coding skill.	Show building **and** coding skills for all 9, and everyone's ideas.
CREATE	DEVELOPING	Squaring and sensors: clear (Accomplished). Forklift: purpose and changes missing. Gyro and PID code: not written.	Every attachment: purpose, what changed, why. Explain the gyro code step by step.
ITERATE	NOT WRITTEN YET	Nothing in the speech yet, but the evidence exists (tuning, 1 m drive, seed basket, full test runs).	Write 2 test stories with numbers: tested, problem, change, tested again.
COMMUNICATE	NOT WRITTEN YET	Lessons and closing not written. The speech has not been practiced out loud.	Write sections 6 and 7, then rehearse. Smile and show pride.

Do these first today

1. **Name the missions** we attempt, in order (section 1).
2. **Finish the wheel story:** what we changed it to and why (section 2).
3. **Forklift:** what it does for Mission 9 and what we changed from the original (section 3).
4. **Write two test stories with numbers** (section 5). This is the biggest gain.
5. **Code:** how Gyro Move, Kp and Ki and yaw turns work (section 4, part 2).
6. **Sections 6 and 7** and the who-says-what table, so all 9 of us speak.

How to reach Exceeds

The judges must write down how we exceeded. Use **numbers**, tell what is **new** compared with the original, and show a **test that led to a change**.

1 Opening and mission strategy

IDENTIFY

WHAT JUDGES WANT

How we decided which missions to try, and which resources and people we learned from. Name at least three resources and say what each gave us.

✎ **Hello, we are team _____ and today we will explain how we designed our robot. Our strategy is to attempt these missions:**

✎ **We chose those missions because ...** (points? location? one attachment for several missions? Mission 11 is worth 30 points)

They got us lots of points and were relatively easy. We looked at all the different missions and ranked them by two things: they needed to be able to get us lots of points, so we have a higher chance of doing better. The other way was how easy it is to achieve the mission, so we can go fast and get more missions done.

✎ **We decided to do them in this order because ...**

We did the missions in that order because we wanted to get from one side to the other just once to save time.

✎ **To learn building and coding we used these resources: (bot-fluencers, conference sessions, SPIKE app, build instructions)**

When building and coding the robot we used bot-fluencers, conference sessions, SPIKE app and build instructions.

✎ **From _____ we learned _____, and we used it for _____.**

✎ **When we got stuck we asked _____ for help with _____.**

DEVELOPING **Review against the rubric (2026-11-12)**

Good: Two clear reasons for picking missions (lots of points, easy to do) and a reason for the order (cross the field once to save time). Resources are named.

Missing: The speech never **names the missions**. The resources are a list with no detail: what did we take from each, and who did we ask for help?

Next: Accomplished: name the missions in order, and give 3 specific resource examples (for example squaring from Programming 201). **Exceeds:** add numbers (points per mission, our 185 point run) and how the plan changed after testing.

2 Choosing and building our robot

IDENTIFY + DESIGN

WHAT JUDGES WANT

A reason for our base robot, not just its name. Facts: we scored 4 designs on 6 criteria about match performance, left cost out on purpose, and HummerOne scored 23 out of 30. See log entries 1, 2, 3, 5 and 9.

✎ **We looked at __ designs: _____. We scored them on __ things: _____.**

We looked at 4 different designs: HummerOne, Comp Bot, GummyBears / Droid Bot M and StarterBot. We scored them on six criteria, all about how the robot performs in a match: archetype match to proven competitive designs, attachment swap speed at base, squaring (flat back and sides), stability under load, maneuverability in tight spaces and drive repeatability (for PID code).

✎ **We left out _____ on purpose because ...**

✎ **The highest score was _____ with __ out of __. We chose it because ...**

✎ **After building it we noticed this problem:** (front wheels? something sticking out?)

After we finished building the HummerOne we noticed the front wheels would slide and create friction, which made it harder to move straight.

✎ **So we changed _____ to _____ because ... (then again, if it changed twice)** (what was wrong with each version?)

DEVELOPING **Review against the rubric (2026-11-12)**

Good: Names all 4 designs and the 6 criteria. An honest problem: the front wheels slid and made it hard to drive straight.

Missing: Why cost was left out, the winning score (23 of 30) and **why we chose HummerOne**, and **what we changed the wheels to and why**.

Next: Accomplished: finish the three blank questions. **Exceeds:** tell the wheel story with all three versions (original, mine cart, ball joint) and the reason for each.

3 Our attachments (part 1)

CREATE

WHAT JUDGES WANT
For each attachment: what it does, what we changed from the original, and why. Do not only say what it is. Say what is new.

✎ **We built ___ attachments. They all attach to the robot's _____ using _____.**

✎ **Our attachments swap on and off in about ___ seconds. We tried _____ pins to make it faster, but ... so we went back to _____ because ...** (log entry 7)

✎ **We rated our attachments from 1 to 5 and chose _____ because ...**

Attachment: _____ **Mission(s):** _____

✎ **It started as a design from:**

✎ **What we changed from the original, and why:**

✎ **A problem we found, and how we fixed it:**

✎ **It is good at ... and not so good at ...**

Our attachments (part 2)

One box per attachment. Choose from: lifting arm with seed bucket, 2-axis arm with hook, forklift, Mission 1 attachment, Mission 11 basket.

Attachment: Forklift **Mission(s):** 9

It started as a design from:

Jack started building the forklift from Coach Zain. Ella and Amelia continued building it.

What we changed from the original, and why:

A problem we found, and how we fixed it:

We made a couple of mistakes, like when Sylvie moved up the gear bar. A problem Ella found was that certain things were built on the wrong side. Sylvie and others fixed this by going back in the instructions and rebuilding certain parts.

It is good at ... and not so good at ...

DEVELOPING **Review against the rubric (2026-11-12)**

Good: An honest story of mistakes and the fix (went back to the instructions and rebuilt), and it names who worked on it.
Missing: What the forklift **does** for Mission 9 (the tree) and how, **what we changed from the original**, what it is good and not so good at, and any test evidence (tipping, "we stabilized it"). Check the source: the draft says Coach Zain, the design log says Forklift1-zoptyk. The lifting arm, 2-axis arm, Mission 1 attachment and Mission 11 basket are not written yet.
Next: Accomplished: every attachment gets its purpose, what we changed and why. **Exceeds:** say what is new and back it up with a test number.

Attachment: **Mission(s):**

It started as a design from:

What we changed from the original, and why:

A problem we found, and how we fixed it:

It is good at ... and not so good at ...

Our attachments (part 3)

Keep going until every attachment we will talk about has a box.

Attachment: _____ **Mission(s):** _____

It started as a design from:

What we changed from the original, and why:

A problem we found, and how we fixed it:

It is good at ... and not so good at ...

Attachment: _____ **Mission(s):** _____

It started as a design from:

What we changed from the original, and why:

A problem we found, and how we fixed it:

It is good at ... and not so good at ...

4 Our code and sensors (part 1)

CREATE

WHAT JUDGES WANT
Explain it step by step, as if the judge has never coded. Say what each sensor does for us.

✎ **We use ___ sensors on our robot: _____. The sensor in the _____ does _____, and we use it to _____.**

We use 2 colour sensors and 1 gyro sensor. The colour sensors sense the colours on the mat. We use them for squaring. The gyro sensor ...

✎ **We added a second color sensor because ...**

The reason behind why we added a 2nd colour sensor is because squaring is when one sensor sees the colour black, it will turn until the other sensor sees black. It will then wiggle till straight. It drives crooked without squaring.

✎ **Squaring up on a line, step by step:** (with one sensor it did not work. With two, what happens when one sees black?)

First, ...

It drives up to the line diagonally and senses the black line.

Then, ...

It turns until the other sensor senses the black line.

Finally, ...

It wiggles until it is straight.

ACCOMPLISHED **Review against the rubric (2026-11-12)**

Good: The squaring steps are clear and in order, and the reason for a second sensor is clear. Sensors are named and each has a job.
Missing: The gyro sentence is unfinished. Where is each sensor mounted? There is no proof yet that squaring works.
Next: Exceeds: finish the gyro sentence, then add before and after proof (for example, how crooked it ended with and without squaring).

4 Our code and sensors (part 2)

CREATE

WHAT JUDGES WANT
Gyro Move, Kp and Ki, gyro turns, the speed ramp and the centimetre conversion. See log entries 8 and 10.

✎ **To drive straight our code uses a block called _____. Every 0.1 seconds the robot checks _____ and if it is off it ...**

✎ **Kp decides _____. Ki decides _____.**

✎ **For turning we use _____ instead of motor degrees because ...** (yaw 90 versus motor 90)

✎ **The robot starts slowly, speeds up and slows at the end because ...**

✎ **We changed the distance block from _____ to _____ because ... We measured that ___ motor degrees is ___ cm by ...**

NOT WRITTEN YET **Review against the rubric (2026-11-12)**
Good: Nothing written yet.
Missing: How Gyro Move, Kp and Ki, yaw turns, the speed ramp and the cm conversion work.
Next: Use log entries 8 and 10 and the Kp and Ki test results in the reference packet. Explain step by step.

5 Testing and improving (part 1)

ITERATE

WHAT JUDGES WANT

This is where teams exceed: **repeated** testing and improvements **because of** the testing. Use numbers. Pick the story you have the best evidence for.

Ideas: ☐ Kp and Ki tuning ☐ 1 m drive, 10 times ☐ seed basket (3, 5, 7 out of 10) ☐ front wheel changes ☐ quick versus axle pins ☐ forklift tipping ☐ squaring with two sensors

✎ **Story 1. We wanted to know ...**

✎ **So we tested by _____, ___ times. The results were: (give numbers)**

✎ **That told us ... So we changed ...**

✎ **After the change we tested again and ...**

NOT WRITTEN YET **Review against the rubric (2026-11-12)**

Good: Nothing written yet, but we **have strong evidence**: Kp and Ki tuning (about 9.6 mm off before, within 1.6 mm after), the 1 m drive 10 times, the seed basket (3, 5, 7 out of 10), and the full test runs on 2026-11-05 (8 runs, best 185 points in 2:50, none under 2:30).

Missing: Every test story needs the pattern: tested, found a problem, changed it, tested again.

Next: Pick two stories and write them with numbers. This is the biggest score gain still available.

5 Testing and improving (part 2)

ITERATE

WHAT JUDGES WANT

A second test story with numbers. Then be honest about what is not finished and say your plan.

✎ **Story 2. We wanted to know ...**

✎ **So we tested by _____, ___ times. The results were: (give numbers)**

✎ **That told us ... So we changed ...**

✎ **After the change we tested again and ...**

✎ **Not finished yet: we still need to _____, and our plan is to _____.** (saying this sounds confident, not weak)

NOT WRITTEN YET **Review against the rubric (2026-11-12)**

Good: Nothing written yet.

Missing: A second test story and what is not finished.

Next: The full test run is a good one: we are 20 seconds over 2:30, so what will we change?

6 What we learned

COMMUNICATE

WHAT JUDGES WANT
A detailed explanation of the process and the lessons we learned. Mistakes that taught us something are good here.

The biggest thing we learned about building a robot is ...

The biggest thing we learned about coding is ...

Something we thought would work but did not was _____. We learned ...

If we were giving advice to a new team we would tell them ...

If we had more time we would ...

NOT WRITTEN YET **Review against the rubric (2026-11-12)**
Good: Nothing written yet.
Missing: Lessons, a mistake we learned from, advice for a new team.
Next: The forklift mistakes (wrong side, gear bar) and the wheel problem are good lessons.

7 Teamwork and closing

DESIGN + COMMUNICATE

WHAT JUDGES WANT

All team members contributed ideas, with building **and** coding skills in all members. Plus pride and enthusiasm, and a Core Value.

Name	One thing I built or designed	One thing I coded, tested or learned
Sylvie		
Brynn		
Jack		
Fedor		
Ira		
Jordan		
Ella		
Alexis		
Amelia		

✎ **Core Value moment (teamwork, discovery, inclusion, innovation, impact or fun): when _____ ...**

✎ **The part of our robot we are proudest of is _____ because ...**

✎ **Thank you for listening. We would love to answer your questions about ...**

NOT WRITTEN YET **Review against the rubric (2026-11-12)**

Good: Nothing written yet. Teamwork already shows in the forklift story: Jack, Ella, Amelia and Sylvie.
Missing: One build thing and one coding or testing thing for all 9 of us.
Next: Fill the table, pick a Core Value moment, and write the closing.

Before we practice

	Checklist
☐	Every blank has a fact, a number or an [ASK: name]
☐	Every choice has a "because"
☐	All 9 teammates speak
☐	At least 2 test stories with numbers
☐	For each attachment: what we changed from the original
☐	Code explained step by step
☐	A lesson from a mistake, and a Core Value
☐	Read out loud, under 5 minutes, without the field and without reading from the page

Questions judges may ask

Jot a short answer or the facts you will use.

1. Why did you choose this base robot?

2. What is the most important change you made, and why?

3. How do you know your robot is accurate?

4. Walk us through how your code works.

5. What did you do when something did not work?

6. Who built each attachment? Who wrote the code?

7. How did you choose which attachment for each mission?

8. What would you change with more time?